Supplementary Information (Online Resource 3)

Article title: How Teams Decide Human Oversight in AI-Powered Systems: A Mixed-Methods Study

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Stakeholders interview demographics

The first 20 interviews spanned a diverse range of domains to capture RE considerations across various contexts. The subsequent 15 interviews focused specifically on high-stakes domains, with an emphasis on participants from the defense sector.

Participants demography:

Participant #	Roles	Years of experience	Domain and context
1	Project Manager, developer	>10	Multidiscipline, computer vision
2	Project Manager	>10	Multidiscipline
3	Project Manager, developer, data scientist, AI expert	>10	Air-Defense
4	Technology management, AI expert, projects entrepreneur	>20	Multidiscipline, AI models and development tools
5	project manager	>5	Automotive industry
6	Project Manager and consultant	>15	Government, critical infrastructures
7	Information Security Specialist, AI expert, developer	>5	Cybersecurity
8	AI expert, developer, startup founder	>10	ChatBot, LLMs
9	Head of development teams, AI expert, product manager	>10	Computer vision
10	Software engineer, AI expert AI startup founder	>10	Retail-tech, LLMs
11	AI expert, data scientist	>10	Transportation and traffic management, computer vision, LLMs
12	AI expert, consultant, regulator	>15	Governance, finance, transportation
13	Technology consultant, head of AI development company, regulator	>15	Strategic business consulting
14	Head of data science team, AI expert	>10	Multidiscipline, governance
15	Customer, data analyst and data scientist	>5	Algo-trading, finance
16	Software engineer, developer	>5	Banking, LLMs
17	AI expert, consultant, data scientist	>10	Multidiscipline
18	Software architect, AI expert, developer	>5	Defense
19	Project Manager, Head of Development teams	>10	Transportation (Airline)
20	Customer, user, business leader	>5	Judicial system

Table 1 – Participants in the first interviews

Participant #	Roles	Years of experience	Domain and context

21	Technology manager, head of department, data expert	>10	Defense
22	Project manager	>5	Defense
23	Head of department, technology manager	>20	Defense
24	Senior Engineer	>15	Defense
25	Team leader, senior engineer, developer, AI expert	>10	Defense
26	Head of department	>15	Defense
27	Deputy Head of R&D	>20	Defense, computer vision
28	Data infrastructure leader	>10	Defense
29	Senior engineer, AI expert	>10	Defense
30	AI technology leader	>15	Defense
31	Deputy Head of R&D	>15	Defense, computer vision
32	Senior engineer	>10	Defense
33	Head of Systems Engineering	>15	Defense
34	System engineer	>10	Defense, computer vision
35	System engineer	>5	Defense

Table 2 – Participants in the second interviews

The extensive representation from the defense sector contributed unique value to the study, particularly given its focus on high-stakes, mission-critical systems where human oversight, safety, and accountability are paramount. The defense domain's rich experience in managing complex, high-risk systems offer important lessons and best practices that are increasingly relevant as AI systems become progressively more embedded in various critical sectors.